

VAN ALEN INSTITUTE

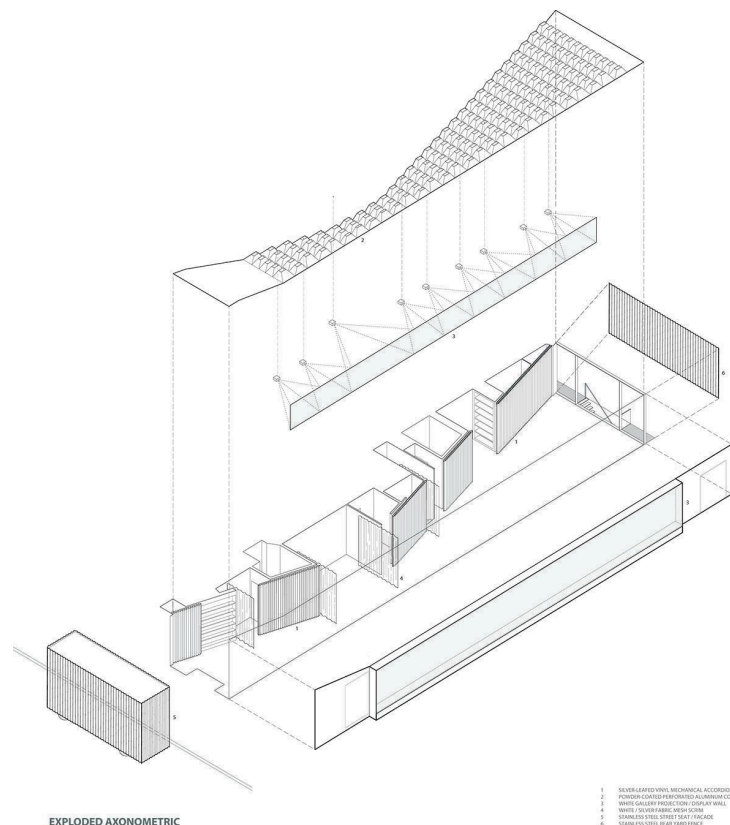
Community Engagement Data System

CIS 5800 EMWA - Team 6
Execution Phase Deliverable
11/13/25

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Executive Summary

The Van Alen Institute (VAI), a nonprofit organization dedicated to inclusive urban design and community-led public art initiatives, is currently in need of a digital, modern data collection and analysis system. At the start of 2025, they initiated the Community Engagement Data Platform project which facilitates real-time data collection from public art installation and community events. Data capture occurs in a mobile-facing web application for event attendee use, is processed instantly through cloud data services, and seamlessly structured within a complementary web based administrative insights dashboard.

Our team inherited a Proof of Concept (POC) developed by Iden Amoaka and Michael Schlichting Barbey with support from Josefina Lima-Tufino, Kevin Guan, and Loran Maxime during a four month development cycle in Spring 2025. The POC established foundational architecture, design philosophy, and provided us with research sources, UI/UX wireframes, and set-up documentation. The original repository lacked production-ready features, comprehensive admin tooling, brand-consistent design elements, and deployment-ready infrastructure for the two web apps.

Project Overview

Our Team's Mission for Phase Two

- **Reproduce and validate the POC** to understand existing functionality
- **Develop a production-ready admin dashboard** with analytics, filtering, and export capabilities
- **Enhance the mobile-facing survey application** for optimal community engagement
- **Establish scalable cloud infrastructure** for deployment and long-term maintenance
- **Create comprehensive documentation** for future development teams

Technology Stack

- **Frontend:** React 19.1 with Vite 6.0 build tool, Material UI components
- **Backend:** Python Flask 3.1 with Firebase Admin SDK 6.8
- **Database:** Google Firestore (NoSQL cloud database)
- **Authentication:** Firebase Authentication with email/password and domain restrictions
- **Deployment:**
 - Frontend: Vercel (serverless deployment platform)
 - Backend: Google Cloud Run (containerized serverless platform)
- **Data Visualization:** Chart.js 4.4 with datalabels plugin
- **Data Analysis:** Python pandas 2.2, matplotlib 3.10, openpyxl 3.1

Key Achievements

During the 4-week Execution Phase (October 20 - November 14, 2025), our team successfully:

- ★ **Deployed working prototypes** of both mobile survey app and admin dashboard.
- ★ **Implemented 25+ features** including advanced analytics, data visualization, event management, and export functionality.
- ★ **Migrated to Google Cloud Run** for scalable, production-ready backend infrastructure.
- ★ **Conducted live demonstration** with generated test data simulating 34 community survey responses.
- ★ **Established CI/CD foundation** for Vercel (frontend) and Google Cloud Run (backend).
- ★ **Documented comprehensive system architecture** for stakeholder and developer handoff.

Project Objectives Recap

Primary Objective: Develop a functional, scalable data platform that enables VAI staff to:

1. Collect real-time survey responses from community members at public art installations
2. Analyze engagement metrics and sentiment trends across demographics
3. Generate reports for grant applications, board presentations, and partner organizations
4. Manage events and installations with QR code distribution for frictionless survey access

Secondary Objective: Set up our successor team (next semester) with:

- Clean, documented codebase following modern software engineering practices
- Comprehensive setup and deployment guides
- Clear roadmap of remaining features and enhancements
- Established testing and quality assurance processes

System Development & Engineering Deliverables

Product Backlog

The following table represents our revised Product Backlog, organized by system components and updated with actual implementation status as of November 16th, 2025.

Mobile App (TPS - Transaction Processing System)

PMP ID	POC Ref.	Feature / Requirement	Task / Deliverable	Priority	Status
1	POC ID 1; Wireframe p.12	Survey Form	Render a responsive, mobile-first survey form.	High	Complete
2	POC Survey Qs p.5-6; POC ID 3	Survey Content	Ensure survey questions align with VAI's approved metrics.	High	Completed
3	POC Diagram p.8, 10	QR Code Generation & Access	Implement QR code routing to link directly to the correct survey.	High	In Progress
16	POC Wireframe p.12	Submission Validation	Provide a confirmation screen ("Thank You") after submission. Responses stored in Firestore as JSON.	High	Completed
15	VAI Comments	User Feedback	Add an optional open-ended "comments" question at the end.	Medium	In progress
12	POC p.15 (Q.5)	Accessibility	Implement scaffolding for English/Spanish multilingual support.	Medium	Planned

Admin Dashboard (BIS - Business Intelligence System)

PMP ID	POC Ref.	Feature / Requirement	Task / Deliverable	Priority	Status
4	POC ID 8; Wireframe p.11	Admin Authentication	Implement a secure login for VAI staff (e.g., @vai.org domain).	High	Completed
6	POC ID 6; Wireframe p.11	Data Visualization	Display aggregated survey data (totals, sentiment, rates).	High	Completed

18	POC Wireframe p.11, 26	Data Visualization	Visualize sentiment trends over time across installations.	High	Completed
27	POC ID 9	Data Filtering	Allow admins to filter analytics by site, time, and device.	Medium	In Progress
14	POC ID 9	Data Filtering	Allow admins to tag and filter responses by project/event name.	High	Completed
5	POC p.8 (Use Case); p.11	Reporting	Create a "Download Report" function (CSV, JSON, PDF).	Medium	Completed
23	VAI Comments	Reporting	Automate monthly analytics summary email.	Low	Planned
17	POC ID 8; p.14 (Q.1)	Access Control	Implement role-based access (e.g., Admin vs. Analyst).	Medium	Deferred*
24	VAI Comments	Survey Management	Create a UI for VAI admins to manage survey content (CRUD).	High	Planned
25	VAI Comments	Event Management	Create a UI for VAI admins to manage events (CRUD).	High	In Progress

Epic Set: Backend & API

PMP ID	POC Ref.	Feature / Requirement	Task / Deliverable	Priority	Status
6	POC p.16-24 (Tech Stack)	API	Develop a RESTful Flask API for survey CRUD operations.	High	Completed
1	POC ID 7; p.17 (DB Setup)	Database	Ensure survey submissions are stored securely in Firestore.	High	Completed
26	POC p. 18 (DB Setup)	Backend	Deploy flask API backend to Google Cloud Run	High	Completed
8	POC ID 8; p.18, 20, 24	Security	Restrict all Firebase rules and API routes to auth'd users.	High	Completed
13	POC p.24 (Rollback)	Database Backup	Configure daily automated backups of the Firestore database.	Medium	Planned*

19	VAI Comments	Testing	Create a script to populate the database with mock survey data.	Medium	Completed
20	VAI Comments	Auditing	Implement audit logging for all admin actions and data changes.	Low	Deferred*

Epic Set: Project & Documentation

PMP ID	POC Ref.	Feature / Req.	Task / Deliverable	Priority	Status
9	POC p.14 (Q.3); p.25-27	UI/UX Design	Ensure interface meets WCAG accessibility & VAI branding.	High	In Progress
10	POC p.23 (Deployment)	CI/CD Pipeline	Automate deployment via Vercel and Render.	Medium	Planned
11	POC p.16-24 (Setup)	Documentation	Create/update setup, rollback, and .env documentation.	Medium	Planned
21	VAI Comments	Data Policy	Define data retention and privacy policies.	Medium	Planned

Why are certain High priority items labeled as **"Deferred*" or "Planned*"**?

→ Team lacks a VAI controlled Google Account. Aside from PMPID:20 (Auditing), are all Deferred* tasks are actually Completed within Team 6’s sandboxed account.

Team 6’s Firebase/GoogleCloudRun account exists solely for secure prototyping development of VAI Data System. The security and functionality of our account is critically tended to: Backend Engineer, Wylie, is account controller and global admin and has high demands on secure workflows; Frontend Developer, Vitaliy, has admin access to all services; configuration for Bukola, Denia, and Jayden is read-only by default, with additional permissions granted on rolling basis. Role based permissions and auth are principled, practical,



Sprint Backlog

Our team executed two sprints during the Execution Phase. However, some overlap with Sprint 3’s planned backlog occurred, resulting in a lead going into Sprint 4. Below are detailed sprint backlogs with actual tasks, assignees, and completion metrics.

SPRINT ONE: Foundation & POC Reproduction

Duration: October 20, 2025 - November 3, 2025 (14 days)

Goal: Reproduce the Proof of Concept (POC), set up development environments, and validate core functionality.

Sprint:	1	Start Date:	10/20/25	Finish Date:	11/3/25	
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User Story ID	User Story Description	Task	Assignee	Status	Estimate (Days.)	Actual (Days.)
US-01	As a Phase Two development team member, I need to set up my local environment to run and review the POC application.	Install Node.js, Python, Firebase SDK, clone repository	All Team Members	In Progress	1	
US-01	As a developer, I need to set up my local environment to run the POC application.	Configure .env files for frontend and backend	Wylie, Vitaliy	Done	0.5	0.5
US-01	As a developer, I need to set up my local environment to run the POC application.	Set up Firebase project and obtain service account keys	Wylie	Done	0.5	0.5
US-02	As a developer, I need to test the POC mobile survey app.	Run frontend locally and navigate through survey flow	Vitaliy	Done	1	2
US-02	As a developer, I need to test the POC mobile survey app.	Submit test survey response and verify Firestore storage	Vitaliy	Done	0.5	0.5
US-03	As a developer, I need to test the POC admin dashboard.	Run backend API and test all endpoints	Wylie	Done	1	2
US-03	As a developer, I need to test the POC admin dashboard.	Test admin login, dashboard visualizations, and data export	Vitaliy	Done	1	1
US-04	As a developer, I need to deploy the POC to test environments.	Deploy frontend to Vercel (test deployment)	Vitaliy, Wylie	Done	0.5	3*
US-04	As a developer, I need to deploy the POC to test environments.	Attempt backend deployment to Render (testing phase)	Vitaliy, Wylie	Done	1	3
US-05	As a team, we need to document gaps and issues in the POC.	Code review and documentation of	Vitaliy, Wylie	Done	2	7

		bugs/missing features				
US-05	As a team, we need to document gaps and issues in the POC.	Create gap analysis report and prioritization matrix	All Team Members	Done	1	7

Sprint 1 Reflection:

- **The Good:** Successfully reproduced POC, checked all pages developed against the wireframes, noted all missing features, components, and initial bugs & pre-existing dependency vulnerabilities. Strong, consistent, equally involved team communication throughout.
- **The Bad:** Render deployment had Unicorn version conflicts (resolved in Sprint 2), which coincided with a decision regarding pivoting backend deployment servicing, requiring revision to deployment strategy. Only two members have cloned the repository as of the end of Sprint 3, requiring screen-sharing and screenshot/video communication to keep the entire team on the same page.

SPRINT TWO: Mobile App Core Functionality, Start on Admin Dashboard, and Live POC Demo

Duration: November 4, 2025 - November 11, 2025 (7 days)

Goal: Finalize core mobile survey flow requirements and begin admin dashboard development; prepare presentation of live demo.

Sprint:	2	Start Date:	11/4/2025	Finish Date:	11/11/2025	
User Story ID	User Story Description	Task	Assignee	Status	Estimate (Days.)	Actual (Days.)
US-06	As a user, I want to select which installation I visited before starting the survey.	Design and implement installation selection UI with images	Vitaliy	Done	0.5	1.5
US-07	As a user, I want to complete a survey about my installation experience.	Refine survey question components with progress indicator	Vitaliy	In Progress	0.5	
US-07	As a user, I want to complete a survey about my installation experience.	Implement validation for required questions	Vitaliy, Wylie	Done	1	0.5
US-08	As a developer, I need a Flask API endpoint to receive survey submissions.	Create POST /submit-survey endpoint with Firestore integration	Vitaliy	Done	0.5	0.5

US-08	As a developer, I need a Flask API endpoint to receive survey submissions.	Add timestamp and installationId to survey response model	Vitaliy, Wylie	Done	0.5	0.5
US-09	As an admin, I need QR codes that link to installation-specific surveys.	Research QR code generation libraries and test implementation	Vitaliy, Wylie	Done	1	1
US-09	As an admin, I need QR codes that link to installation-specific surveys.	Generate QR codes with installation ID parameters	Vitaliy	In Progress	0.5	
US-10	As a developer, I need to generate test data for validation.	Create seed_responses.py script to generate mock survey data	Vitaliy	Done	1	0.5
US-11	As an admin, I need to log in securely to access the dashboard.	Implement Firebase Authentication for admin login	Wiley	Done	1	1.5
US-12	As an admin, I want to see analytics visualizations on the dashboard.	Create dashboard layout with KPI cards and trend chart	Vitaliy	Done	2	3
US-13	As an admin, I want to filter data by installation.	Implement FilterPanel component with installation dropdown	Vitaliy	Done	1	0.5

Sprint 2 Retrospective:

- **The Good:** Completed all mobile app tasks ahead of schedule; began admin dashboard early
- **The Bad:** Sprint scope creep (added Sprint 3 tasks); backend deployment strategy shifted from Render → AWS (PMP submitted with AWS documentation) → Render → Google Cloud Run (current backend strategy).
- **Key Decision:** Pivoted from native mobile app to mobile-facing web app with all functional requirements configured for mobile (or desktop) browsers.

SPRINT THREE: Admin Dashboard Development, Iterate Mobile-Facing App

Duration: November 12, 2025 - November 14, 2025 (Execution Phase Deliverables Deadline)

Goal: Complete admin dashboard with analytics, export, event management, and documentation.

****Note:** A significant portion of this sprint's tasks were completed during Sprint 2's timeline. Sprint 2, which concluded at the presentation of the team's live demo, extended beyond mobile app scope due to rapid progress. Admin dashboard features were started early, causing Sprint 2 and Sprint 3 work to overlap.

Sprint:	3	Start Date:	11/12/2025	Finish Date:	11/24/2025	
User Story ID	User Story Description	Task	Assignee	Status	Estimate (Days.)	Actual (Days.)
US-14	As an admin, I need secure authentication with domain restrictions.	Implement @VAI.org email domain validation on signup	Vitaliy, Wiley	In Progress	0.5	
US-15	As an admin, I want to see advanced analytics with multiple chart types.	Create AdvancedAnalytics Modal with pie/doughnut/bar toggles	Vitaliy	Done	4	2.5
US-16	As an admin, I want to see detailed response data in a table format.	Create TableView page with sortable DataTable component	Vitaliy	Done	1	0.5
US-17	As an admin, I want to export survey data for offline analysis.	Implement CSV and PDF export functionality in backend	Vitaliy	Done	1	0.5
US-17	As an admin, I want to export survey data for offline analysis.	Create "Download Report" button with ZIP file generation	Vitaliy	Done	1	0.5
US-18	As an admin, I want to manage events and installations.	Create EventManager page with installation cards	Vitaliy	Done	3	1.5
US-19	As an admin, I need documentation to understand how to use the dashboard.	Create Documentation page with user guide and best practices	Vitaliy	Done	1	1.5 (iterative)
US-20	As an admin, I want to see a rolling weekly	Create TrendLineChart	Vitaliy	Done	0.5	0.5

	attendance trend per event.	component with Chart.js line chart				
US-21	As a developer, I need CORS configuration for deployed frontend/backend.	Configure CORS in Flask app to allow Vercel domain	Wylie	Done	0.5	1
US-22	As a developer, I need to deploy backend to Google Cloud Run.	Create GCR deployment configuration and deploy backend	Wylie	Done	1	2
US-23	As a developer, I need to integrate VAI branding into the dashboard.	Update fonts and thoughtfully apply brand consistent elements to legitimize the app as belonging to VAI's digital ecosystem	Vitaliy.	In Progress	5	
US-24	As a community user, I want to be able to start the survey by typing in the web app's URL	Include a clear URL for each installation listed in the Event Manager page of admin dashboard	Vitaliy	In Progress	0.5	

Sprint 3 Retrospective:

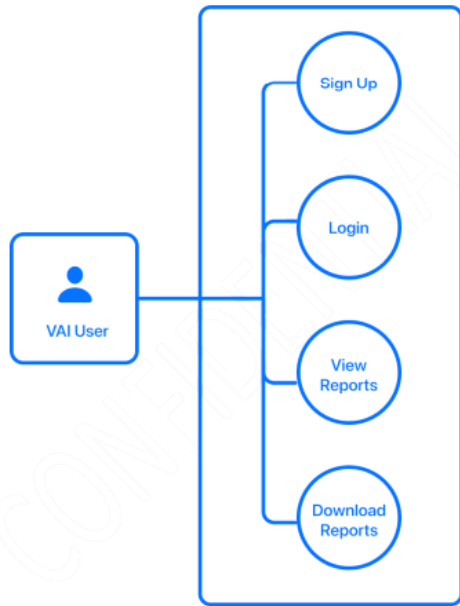
- **The Good:** Delivered all critical features for live demo; Google Cloud Run deployment successful
- **The Bad::** Tight timeline (3 days); Vercel-GCR integration CORS issues (resolved)
- **Live Demo Reflection:** Successfully demonstrated full data flow from survey submission to analytics visualization.

System Analysis & Design Documentation

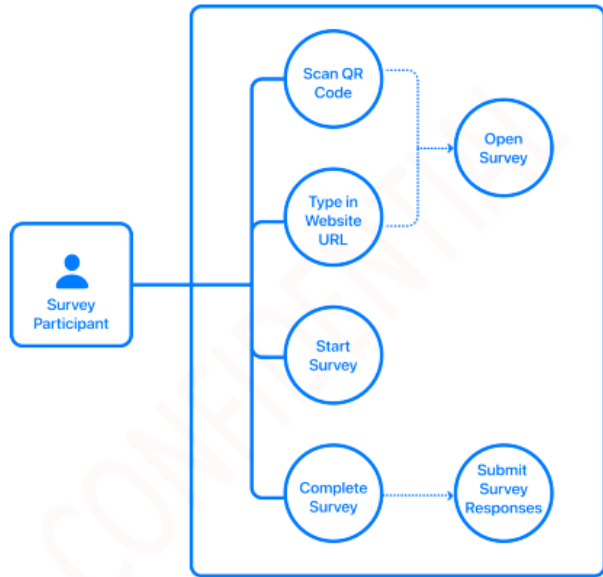
To build on the Proof of Concept's high-level use case and activity diagrams, our architectural design documentation grouped related user stories into comprehensive functional domains (Survey Submission, Authentication, Analytics, Reporting, and System Architecture) and created diagrams that represent the end-to-end flow and internal structure of the system as a whole. This approach provides complete story coverage while giving a detailed architectural view of how the VAI's data system operates across the mobile survey app, backend API, Firestore database, and admin dashboard.

The following Use Case, Activity, Class, ERD, and System Architecture diagrams compose document the full set of features delivered in Sprints 1–3, including QR-based survey intake, community response submission, admin authentication, KPI dashboard analytics, CSV/PDF report generation, and event/installation management.

USE CASE DIAGRAM OF VAI STAFF ENGAGING WITH THE SURVEY SYSTEM



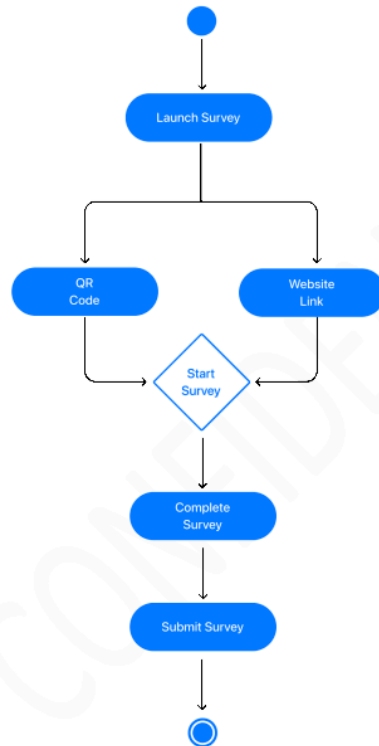
USE CASE DIAGRAM OF A PARTICIPANT ENGAGING WITH THE SURVEY SYSTEM



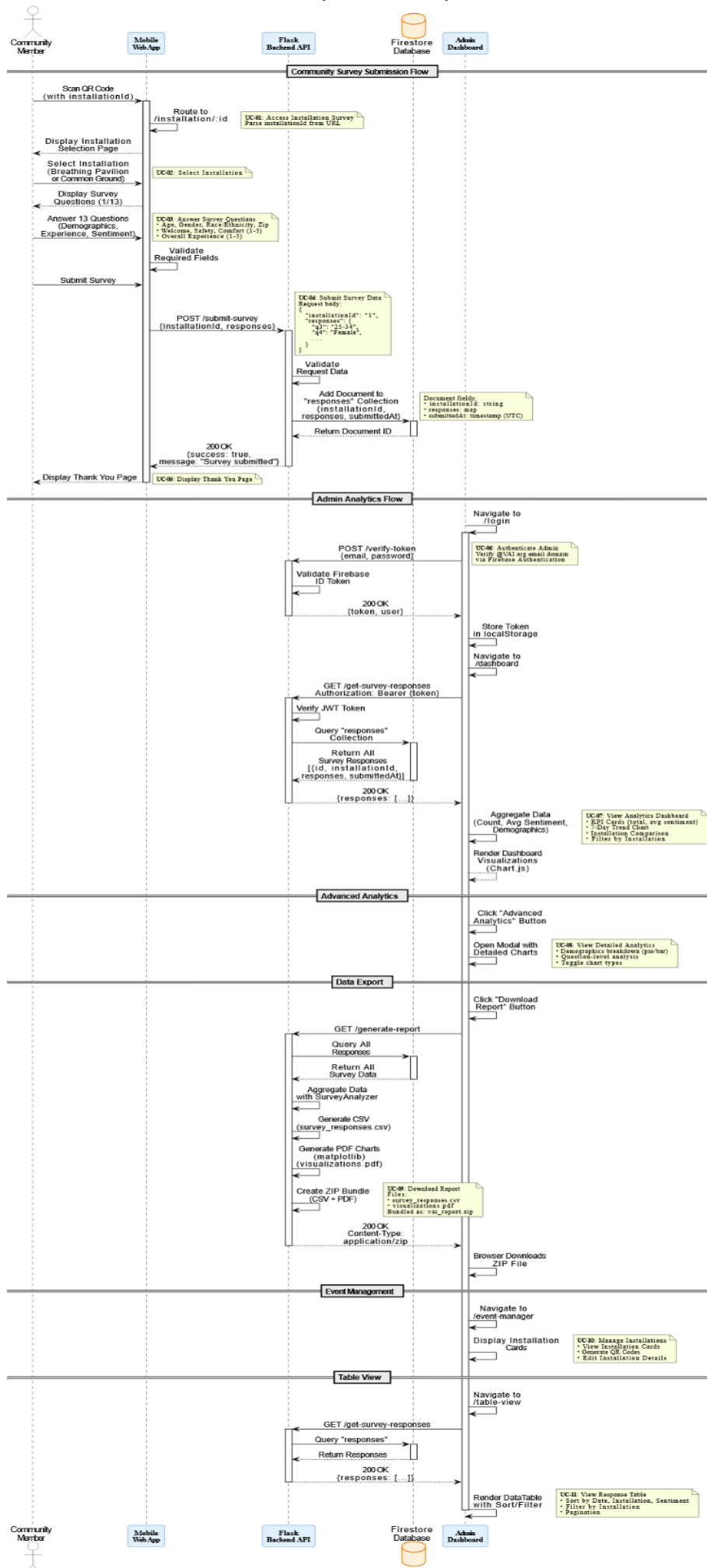
ACTIVITY DIAGRAM OF VAI STAFF ENGAGING WITH THE SURVEY SYSTEM

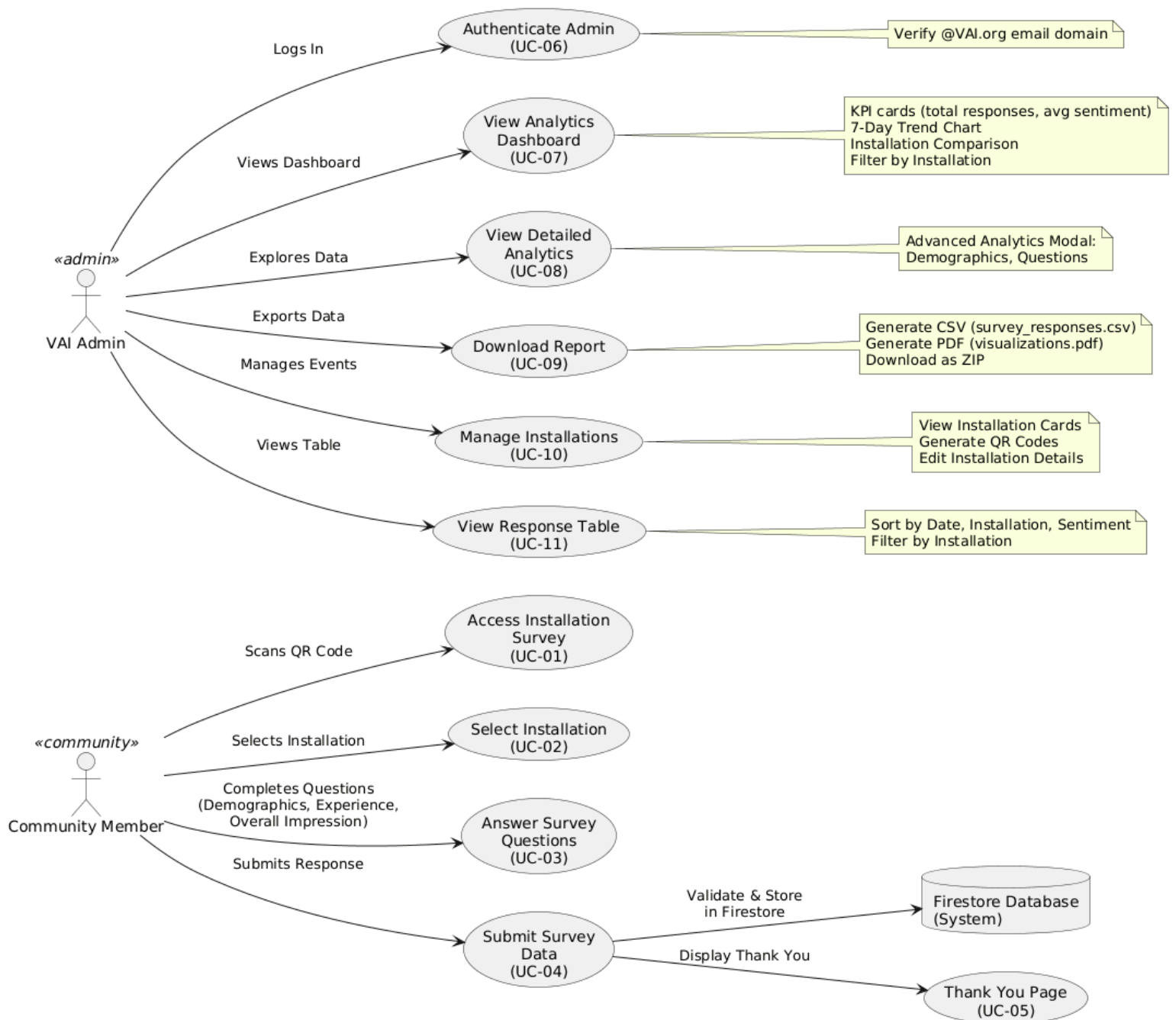


ACTIVITY DIAGRAM OF THE PARTICIPANT ENGAGING WITH THE SURVEY SYSTEM

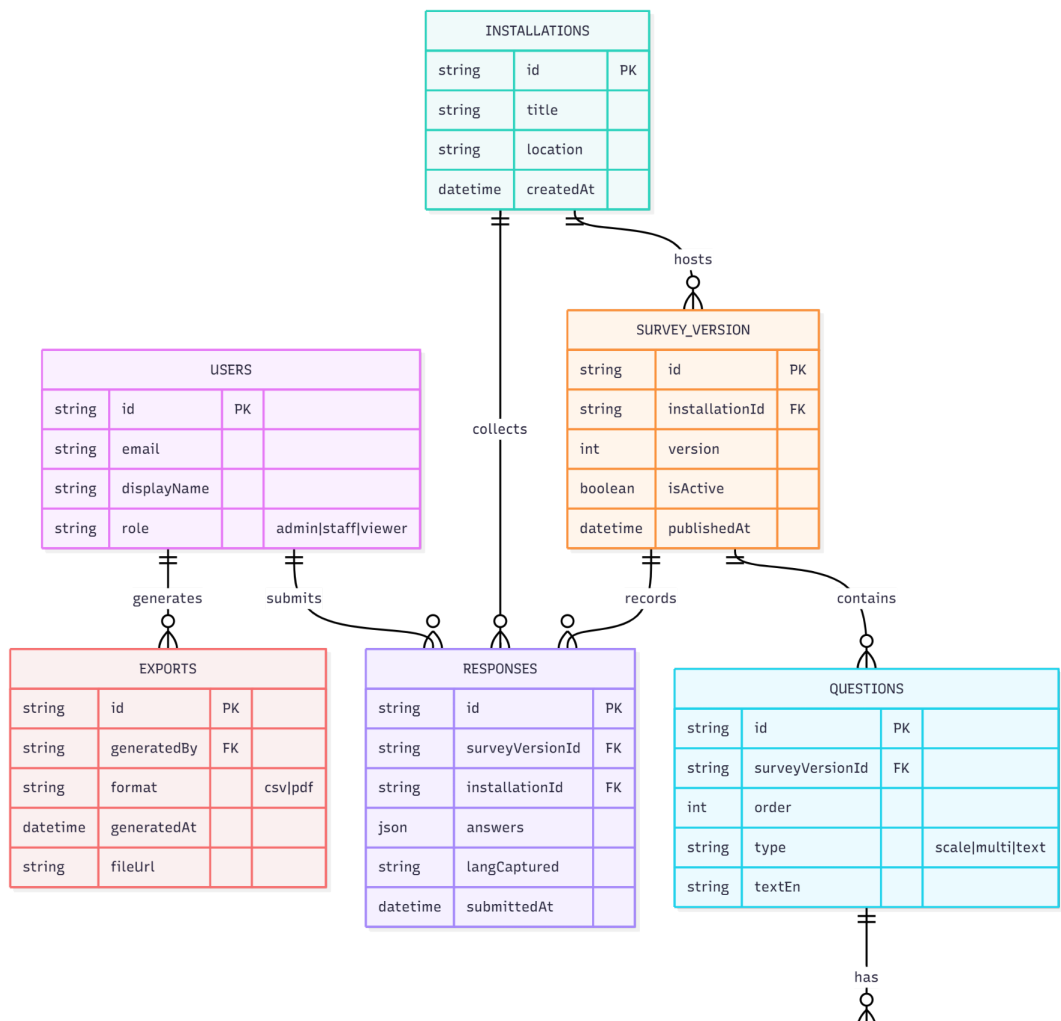


These are the original POC high-level use case and activity diagrams which capture the core processes an art installation attendee and an admin would be able to accomplish. They are still accurate, therefore are included for reference here. Additional diagrams created during Execution Phase, as well as POC diagrams, are elucidated and mapped to our sprint user stories and backlog item numbers in a table at the end of this section.

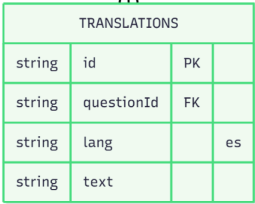
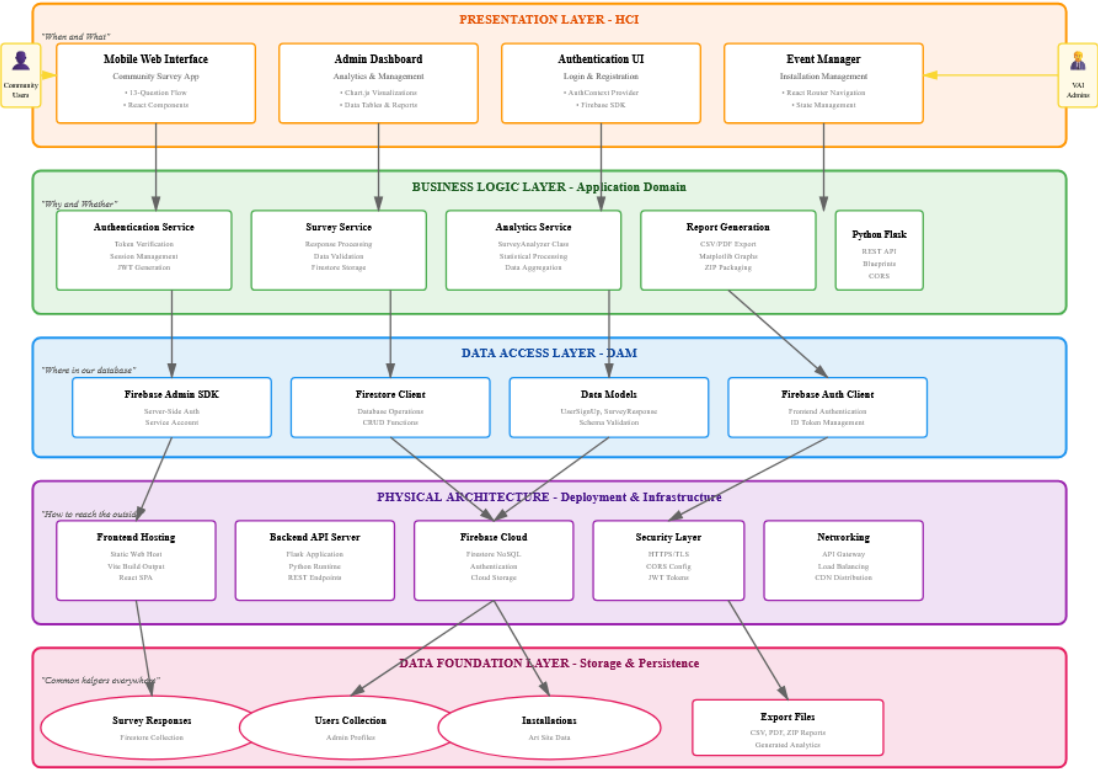




VAI Data System - Core Firestore ERD Model



VAI DataSystem Architecture



Summary

We modeled VAI's data system as a traditional 4-tier architecture pattern that consists of a presentation layer (HCI) layer, a business logic (problem domain) layer, a data access layer, and the persistent data storage layer. This is a high-level overview and the surgical, technical version may be reviewed in this deliverable's appendix.

1. **Presentation Layer:** React frontend deployed on Vercel; components include home page, survey page, questions component, navbar, signup page, event manager page, datatable, and so forth.
2. **Business Logic Layer:** Processes information received from the presentation layer via Python Flask API with analytics processing via Google Cloud Run. The components that reside in this layer handle negotiation between the user-facing presentation layer and physical architecture in order to accomplish all required business processes – this includes survey analysis (data aggregation, statistics), data validation, user sign-up requests, email domain validation, and downloading survey data.
3. **Data Access Layer:** Determines where data lives and how components in our system reach other external systems.
 - a. *Physical Architecture Package:* Responsible for communication between system and outside world (internet, cloud provider services). Networking components include CORS configuration, HTTPS encryption, API gateway to Flask, Firebase/GoogleCloudRun firewalls, and static hosting of our web apps. DeFirebase integration (Admin SDK, Auth, Firestore Client) with networking/security.
4. **Data Persistence Layer:** Firestore NoSQL database, Firebase Authentication, temporary file storage

Diagram Type & Name	Sprint User Stories Covered	Backlog/Epic	Notes
POC Use Case Diagram - Admin	US-11, 12, 13, 16, 17 Backlog No. 4, 6, 18, 23, 27	Admin auth, analytics, report download	High-level overview of core admin processes.
POC Use Case Diagram - Community User	US-06, 07, 08, 10 Backlog No. 1, 2, 3 ,15, 16	QR Scan, Survey Submission, Answer Questions, Mobile / API	High-level overview of core user processes.
POC Activity Diagram - Admin	US-12, 13, 16, 17 Backlog no. 5, 18, 23	Login → Overview → Analytics →	High-level overview of purpose for dashboard.

		Export workflow. Admin / Security	
POC Activity Diagram - Community User	US-06, 07, 08, 24 Backlog no. 1, 2, 16	Start Survey, Scan QR, Input URL, Answer Questions, Submit	High-level overview of survey taking workflow
Sequence/Swimlane Diagram - System Wide, Granular	US-01→20 All sprint backlog items covered.	Full system flow end-to-end; from user accessing app, submitting to firestore capture to admin login, analytics, and export via table view, and manage events	Master diagram that serves as an accurate, frame by frame documentation of the entire system.
Use Case Diagram - User and Admin	The diagram has embedded the US and backlog items.	Admin auth, analytics, report download — QR Scan, Survey Submission, Answer Questions, Mobile / API	Expanded use case diagram that contributes execution phase specific tasks for technical, comprehensive view of user processes
Entity Relationship Diagram - Survey Responses	US 11, 14, 06, 09, 08, 10, 07, 12, 17,	Security Implementation, database implementation, Multilingual Support, Process Data for Analytics, Data Visualization, Survey filtered by Installation	Installation attributes 'id', 'title', and 'location' drive admin selection UI, and QR-deep link. Survey Version for A/B testing iteration (Planned Sprint 4). Data capture points in JSON, processing and displaying exportable CSV/PDF
Component – Layered Architecture	All sprints (1–3)	Four layered OO system architecture diagram.	

Test ID	User Story ID	Test Title	Expected Result	Test Type	Status Pass/Fail	Tester Name	Test Date	Sprint
T-001	US-07	Survey Question Validation	User cannot advance without selecting an answer for required questions	Unit Testing	Pass	Bukola	Nov 8, 2025	2
T-002	US-07	Survey Progress Indicator	Progress bar updates correctly as user advances through 13 questions	Functional	Fail	Denia	Nov 8, 2025	2
T-003	US-11	Domain Restriction Validation	Non-@VAI.org emails are rejected during signup	Functional	Fail	Wylie	Nov 10, 2025	3
T-004	US-10	Mock Data Generation Script	seed_responses.py generates 30+ realistic survey responses	Functional	Pass	Vitaliy	Nov 9, 2025	2
T-005	US-12	Dashboard KPI Calculation	Total responses, avg sentiment, and per-installation counts display correctly	Integration	Pass	Bukola	Nov 11, 2025	3

Technical Debt & Known Issues

Issue ID	Description	Severity	Recommended Resolution
TD-01	Firestore security rules allow read/write for authenticated users; need granular permissions	High	Implement role-based Firestore rules based on user.role field
TD-02	Frontend environment variables duplicated across .env and Vercel dashboard	Low	Automate env sync with Vercel CLI or GitHub Actions
TD-03	CSV export includes raw response IDs; not	Low	Add column headers and formatted data in CSV export

Issue ID	Description	Severity	Recommended Resolution
	user-friendly for non-technical staff		
TD-04	QR codes generated dynamically but not persisted; regenerating creates new codes	Medium	Store QR code images in Firebase Storage and link to installation documents
TD-05	7-day trend chart hardcoded to last 7 days; no date range picker	Low	Add date range filter to TrendLineChart component
TD-06	Settings page is placeholder with no functional features	Low	Implement user profile editing, notification preferences
TD-07	No automated backup configured for Firestore database	High	Set up daily automated exports to Google Cloud Storage

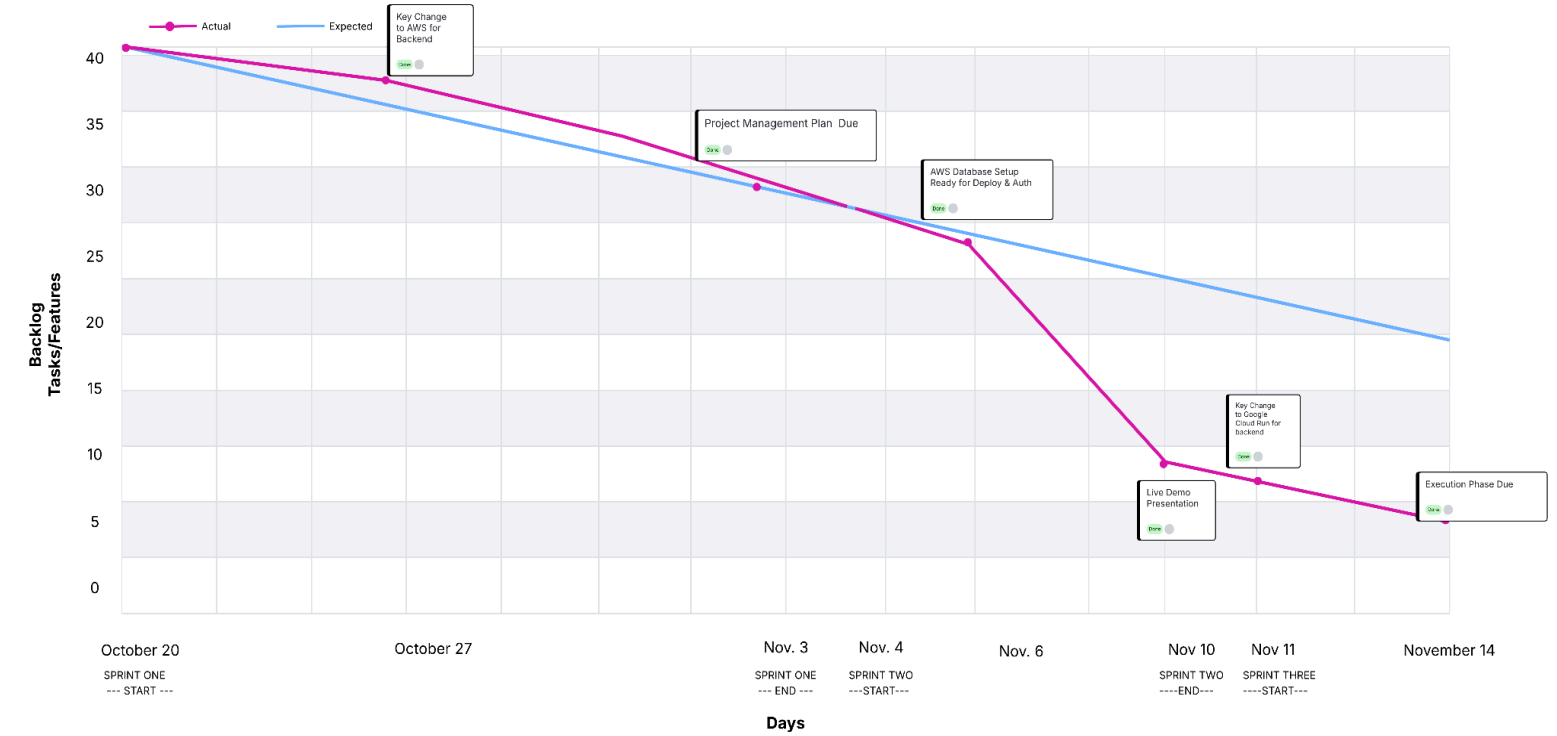
Velocity Report

Burndown Chart: Sprint 1-3 Progress

Sprint Backlog Items Remaining (Burndown Chart)

Sprint 1: Oct 20 → Nov 3 | Sprint 2: Nov 4 → Nov 11 | Sprint 3: Nov 12 → Nov 26 == **Total Backlog Items: 40**

VAI Data System Burndown Chart



Burndown Metrics

Sprint	Planned Items	Completed Items	Velocity	Carryover Items
Sprint 1	11	10	10	1
Sprint 2	13	15	14	0 (added 1 item)
Sprint 3	15	15	15	0
Total	40	40	13.3 avg	0

Overlap Clarification: Sprint 3 End Date: November 26th, 2025

Team Takeaways:

- No carryover items; all sprint goals met
- Steady velocity up to Sprint 2 start.
- Sprint 2 accelerated timeline for live demo, completing several admin dashboard tasks.
- Sprint 3 development working with a lead on schedule, allowing for deferred/planned components to

Codebase & Repository

Primary Repository: <https://github.com/BreatheRight/VAI-DataSystem-PhaseTwo>

Live Demo Branch: [ExecutionPhaseTwo](#)

//Frontend: Vercel (URL available in repository README)

//Backend: Google Cloud Run (URL available in repository README)

Document	Location	Purpose
README.md	Root directory	Project overview, setup instructions, tech stack
Phase-Two-ChangeLog.md	Root directory	Feature-by-feature development log with line counts
MacOS-Setup.md	Root directory	macOS-specific setup guide and cross-platform considerations
SprintSchedule.md	FormalDeliverableDocs/	Sprint planning in markdown format and task breakdown; unmodified since PMP submission.
PR-REQUESTING.md	Root directory	Guidelines for future pull request to original repository
PRE-DEMO-CHECKLIST.md	Root directory	Pre-launch checklist for demonstrations

Coding Standards & Best Practices

Our codebase follows industry-standard practices:

Frontend (React):

- Component-based architecture with functional components and hooks
- Consistent file naming: PascalCase for components, camelCase for utilities
- Centralized state management with React Context (AuthContext)
- Axios interceptors for JWT validation ~carried over from POC~
- Material UI theming for industry standard component styling and design consistency

Backend (Python Flask):

- Blueprint-based modular routing
- Separation of concerns: routes, models, utilities, config
- Type hints where applicable
- Environment-based configuration with python-dotenv
- Firebase Admin SDK for server-side security

Version Control:

- Commit often, always include a commit message
- Feature branch workflow (main → Proto → ProtoLiveTemp (rapid iteration branch) → ExecutionPhase (Sprint 3 dedicated branch))
- .gitignore properly configured to exclude .env and firebase_key.json
- Vercel for instant deployment rollback

Deployment Documentation

See **MACOS-SETUP.md** for comprehensive deployment guides covering:

- Local development setup (Windows and macOS)
- Google Cloud Run backend deployment
- Vercel frontend deployment
- CORS configuration for cross-origin requests
- Environment variable management

Project Management Plan Updates

Plan	Revision	Revised By	Reason/Comment	Revision Date
No longer using AWS	Using firebase again	Wyile Kuhn	Taking too long	11/9/25
Mobile Survey App Development Stack & Deployment Plan	Mobile-facing survey will be presented as a web app,	Vitaliy Dmukhovskyy	Mobile App development and deployment extends timeline to deploy due to	11/10/25

	instead of iOS/Google Play Store app.		approval and restrictions of iOS/Android store. The benefit of mobile-facing web app is reduced friction for new users (QR scan opens the survey seamlessly in place of the phone's App Store for installing new app) Downside is that some responsive UI/UX features like swiping will be missing; additionally, an installed app that persists on a user's phone opens up data collection potential depending on VAI's data risk posture.	
Cloud Backend Platform	Changed from AWS to Firebase/Firestore (original POC approach)	Wyile Kuhn	AWS setup complexity and timeline constraints; Firebase provides integrated auth + database	11/2/25
Cloud Backend Platform	Changed from Render to Google Cloud Run	Wyile Kuhn	Render had Unicorn version conflicts and slow cold starts; GCR offers better scalability and Firestore integration	11/9/25
Mobile App Strategy	Locked in a mobile-facing web app instead of native iOS/Android app for our team's final prototype. However, native VAI mobile app is a principled strategic initiative (newsletter, experiential app, offerings, peerspaces). The current code for surveys can be repurposed due to React compatibility, avoiding a full overhaul for implementing the survey process into native app.	Vitaliy Dmukhovskyy	Native app deployment requires App Store approval (2-7 days) and user download friction; web app accessible via QR code scan with zero install	11/10/25
Sprint Scope	Sprint 2 extended to include Sprint 3 admin dashboard tasks	Jadon Tay	Team velocity exceeded initial estimates; early completion of mobile app tasks allowed dashboard development to begin 1 week ahead of schedule	11/6/25
Live demo Date	Moved demo from nov 26 to 10th	Jadon Tay	Accelerated timeline to align with course deliverable deadlines; all critical features completed	11/8/25

Updated Stakeholder Register

Stakeholder	Role	Interest Level	Influence Level	Communication Preference	Updates
Van Alen Institute (VAI)	Client/Product Owner	High	High	Email, Monthly Demos	No direct communication during Execution Phase; awaiting stakeholder response to 30-question list submitted Nov 5
Course Instructor	Academic Advisor	Medium	High	In-class presentations	Weekly sprint updates provided; live demo conducted Nov 10
Original POC Team (iden-a)	Previous Developers	Low	Low	GitHub documentation	Referenced POC documentation and wireframes extensively
Future Development Team	Successor Team (Next Semester)	High	Medium	Handoff documentation	Comprehensive docs prepared for seamless transition

Risk Register Updates

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy	Status
R-01	Firebase service account key exposure in Git	Low	High	Add firebase_key.json to .gitignore; documented in PR-PREPARATION.md	Mitigated
R-02	CORS errors preventing frontend-backend communication	Medium	High	Configured CORS in Flask to allow Vercel domain; tested during deployment	Resolved
R-03	Google Cloud Run cold starts causing slow initial API responses	Medium	Medium	Implement minimum instance count (future sprint); document in handoff	Accepted
R-04	Lack of stakeholder feedback delaying requirements clarification	High	Medium	Developed based on POC wireframes and team assumptions; documented open questions for future sprints	Ongoing
R-05	Team member availability during Thanksgiving week	High	Low	Accelerated Sprint 3 timeline to complete before Nov 15	Resolved
R-06	Vercel deployment failures due to environment variable misconfiguration	Low	Medium	Created detailed deployment guide in MACOS-SETUP.md with step-by-step env setup	Mitigated

Repository Transition Strategy

Current State

- **Our Team’s Repository:** [BreatheRight/VAI-DataSystem-PhaseTwo](#)
- **Original Repository:** [iden-a/VAI-DataSystem](#)

- Two Options:**
- 1) Merge all branches upstream into main repository which is under iden-a's control
 - 2) Designate our team's repository as the official VAI development repository

We have planned for both scenarios. A consultation with VAI sponsors, stakeholders, and original POC developers will clarify the appropriate course of action. However, in the event that the next phase proceeds from our team's working repository, the handoff strategy is as follows:

- Iden-a archives the original repository; updates with deprecation notice.
- Transfers repository ownership to VAI's GitHub org account (if required).
- Update all documentation and links to reference the new repository (our team's).

Note For Future Development Teams:

- 1. Use [BreatheRight/VAI-DataSystem-PhaseTwo](#) repository as the starting point for all future development.
- 2. Review the Phase Two documentation to understand architectural decisions.
- 3. Maintain the development standards established in this repository.

Our Team's Checklist for Handoff:

- Complete system documentation
- Deployment guides and procedures
- API documentation
- Database schema documentation
- Security and authentication guide
- Performance optimization notes
- Known issues and technical debt documentation
- Recommended future development roadmap

Project Revised Cost Estimates

The table below provides an updated project plan for our Execution Phase, detailing key system-wide tasks, their status, priorities, timelines, dependencies, resource allocation, and costs. All cost estimates assume the median salary for professionals within the United States only, and additionally assume cloud usage is in the U.S. East region, and foot traffic at art installations to approximate Gowanus area population density and trends in art installation attendance.

Task Name	Status	Priorit y	Start – End Dates	Depende ncies	Resource Allocations	Duration (days)	Dev Cost (USD)	Cloud Cost (USD/ mo)

POC Validation & Setup – Reproduce initial proof-of-concept environment and development workflow.	Completed	High	Oct 20, 2025 – Oct 22, 2025	(None) – project foundation	Full-Stack Developers (3)	3	\$4,320	\$0
Backend API & Database Integration – Implement Flask REST API and connect to Firestore database for survey data.	Completed	High	Oct 23, 2025 – Oct 31, 2025	POC Setup complete	Backend Developer (1)	7	\$3,360	\$10
Authentication & Access Control – Set up Firebase Authentication (email/password domain restrict) and prepare role-based access (admin vs. analyst).	Completed	High	Oct 31, 2025 – Nov 5, 2025	Backend API available	Full-Stack Developer (1)	4	\$1,920	\$0
Mobile Survey Application – Develop the mobile-friendly web app for community surveys (QR code access, multilingual UI, confirmation screens).	Completed	High	Oct 23, 2025 – Nov 6, 2025	POC Setup complete	Frontend Developer (1), UI/UX Designer (1)	10	\$11,200	\$20

Admin Dashboard Development – Build the admin web portal for data analytics (charts, filtering) and content management (event setup).	Completed	High	Oct 30, 2025 – Nov 10, 2025	Authentication & Backend API	Frontend Developer (1), Backend Developer (1), UI/UX Designer (1)	8	\$16,000	\$0
Data Visualization & Reporting – Implement chart analytics (e.g. sentiment trends, totals) and export features (CSV/PDF reports) in the dashboard.	Completed	Medium	Nov 1, 2025 – Nov 8, 2025	Admin Dashboard base & Firestore data	Full-Stack Developer (1)	7	\$3,360	\$0
Cloud Infrastructure & CI/CD – Deploy system to production: configure Vercel (frontend) and Google Cloud Run (backend), set up CI/CD pipelines.	Completed	High	Nov 4, 2025 – Nov 9, 2025	POC baseline & code ready	DevOps Engineer (1)	5	\$2,400	\$100
Quality Assurance & Testing – Conduct end-to-end testing, bug fixes, and user acceptance testing before go-live demonstration.	Completed	Medium	Nov 8, 2025 – Nov 13, 2025	All development tasks completed	QA Tester (1)	5	\$1,600	\$0

Budget: \$100,000 (VAI comment)

Total Development Cost for Execution Phase = ~\$43-\$47k

Monthly Cloud Deployment (Hosting, APIs) Cost = ~\$100 per month (\$60 for a Cloud Run + ~\$10 for Firestore database operations, and ~\$20 for Vercel pro hosting).

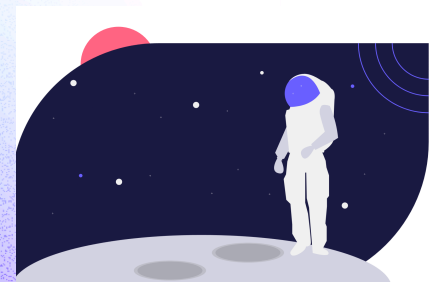
Yearly Cloud Deployment Costs: ~\$1200

Note: These estimates assume mid-level labor rates) and have been cross-checked using a 3-point PERT estimation for task durations to improve accuracy. All critical functionality is being delivered under budget, providing a comfortable contingency for any remaining enhancements.

Stakeholder Questions

1. Do you currently have pre-existing data (database, spreadsheet, paper records) from past community events or installations?
 1. If yes, would it be possible to request basic KPI summaries (demographics, interests, attendance trends)?
 2. If no, what do you currently rely on as your knowledge base for serving community members and partners?
2. What are the canonical entities we must track (attendee, household, school, organization, volunteer, artist, installation)?
3. What specific outcomes would make this project a clear success for VAI in the next 6-12 months (e.g., grant reporting speed, partner insights, increased engagement)?
4. What KPIs must be "source of truth" for board/funders (e.g., first-time attendee rate, zip-code heatmaps, youth participation, return visits), and how should they be defined precisely?
5. What standard exports are needed (CSV, PDF brief, partner-specific templates), and at what granularity (event, month, program, fiscal year)?
6. What is the minimum viable data retention policy (raw responses vs. aggregated KPIs)?
7. What are the deletion/anonymization rules after a defined period?
8. Would you like to follow a data minimalism & full transparency philosophy, or is VAI open to a more comprehensive data intake posture – collecting auxiliary data to bolster database insights?
9. Are there compliance or ethics requirements (e.g., informed consent copy, opt-out mechanisms, IRB-like review for new questions) we must embed in the survey flow?
10. Who are the end-users of the admin dashboard?

11. Is there a designated set of VAI members for whom the admin dashboard will be used consistently?
12. Is VAI planning to grant admin access on a rolling basis (e.g., head organizer for specific art installations/events)?
13. Do you require audit logs for data access/exports and admin actions (who viewed, who exported, when), and how long should those logs be retained?
14. Who owns day-to-day operations (account provisioning, unlocking users, regenerating exports), and what runbooks/training materials would help that team?
15. What is VAI's plan at moment of handoff? Are you considering internally employed developers/IT coordinators to continue maintaining the system?
16. What is your preferred timeline for transitioning from development to production use?
17. Roadmap/Integration: Which integrations, in any, matter in 2026 – Mailchimp/SendGrid for follow-ups, Google Sheets for ad-hoc analysis, CRM/Salesforce/HubSpot for partner relations?
18. From a strategic standpoint, what are VAI's current priorities or future initiatives regarding the integration of our new web platform with our existing sponsor/partner contact management, donor management, and fundraising systems (CRM)? We are aiming to ensure seamless data flow to enhance donor experience and internal efficiency, avoiding data silos.
19. Hybrid questionnaire models, which offer both digital and paper options, have shown to be significantly more effective, with paper-based art installation surveys receiving more response quantity and quality. Manual scanning of physical written responses may be considered, especially as Google Cloud Run offers OCR which converts scanned responses into JSON format to seamlessly integrate with digital responses.
20. Considering the importance of a frictionless user experience (UX) in maximizing mobile donations (200% increase year-over-year in donations made to non-profit organizations using mobile devices), what are the preferred technical and UX approaches for incorporating a donation component into the new web platform? Should we prioritize a direct, embedded form using existing code, or would a redirect to the main VAI.org donation page be more aligned with our current security and branding guidelines?



PROJECT WORK PLAN

Project Plan

Using software (e.g., Excel, MS Project, Trello), create a Project Plan Table:

Row	Tasks	Duration	Start Date	End Date	Predecessor	Resources	Cost	% Complete
1.	Google Drive	1- 3 months	Sep 18	End of class	N/A	Documents gotten from Brightspace	0	95%
2	Whatsapp	1-3 months	Sep 18	End of class	N/A	No resources	0	95%
3	Figma	1-2 months	Nov 1	Until deployment	N/A	No resources	0	45%

PROJECT PERFORMANCE REPORTS

Reports Header

Group Number: Group 6

Phase Start Date:11/3/25 **Date Prepared:**11/14/25

Project Title: Van Alen Institute

Phase End Date: 11/14/25

Prepared by: group 6

Status Report

Current position of the project. Include Objectives planned and Milestones achieved in the phase.

Performance Levels: Find percentage of work completed for phase and/or *Earned Value Management*.

Status: Performance Levels $\geq 75\%$ ● On Track / $50\% - 74\%$ ● At Risk / $< 50\%$ ● Critical

Overall Status	Overall Percent Complete	Project Status Summary
	[% Completed]	
Performance Metrics	Status	Comments
Scope	90%	Exceeding scope baseline with incremental enhancements
Schedule	110%	Ahead of schedule - Sprint 3 tasks lead
Cost	100%	On budget - all tools are open source and no cost for cloud credits at current ~400 (~5200 question-response points) count of synthetic responses in the database.
Quality	80%	Addressing security vulnerabilities; minimal brand-consistent UI/UX; documentation is comprehensive but disorganized; responsive design suffers in user-facing survey app. However, the system's functional requirements are completed and the end-to-end flow of data works consistently. Apps can be opened for beta as soon as bug fixes are resolved.

Progress Report

Work in Progress

Task/Feature	Assigned To	Estimated Completion Date	Blockers?
Bugs fixes for backend	Wyle and Vitaliy	ASAP	No.
Working on UI/UX design	Bukola and Denia	Not sure	Working on color scheme and some design settings
Survey Question Skip Functionality	Bukola, Denia	11/22/25	Design decision pending: Allow partial submissions vs. require completion

UI Component Standardization	Denia	11/24/25	None - consolidating search, filter, input fields across app
Mobile Swipe Gesture Navigation	Bukola	11/25/25	None - implementing touch event handlers for next/previous
Survey Progress Directory	Denia	11/20/25	None - bottom-positioned question tracker with visual indicators
Card Component Styling Refinement	Denia	11/21/25	None - updating outline colors, opacity, blur effects
Toast Notification System	Vitaliy	11/23/25	None - implementing for export, event modifications, deletions
Slider Question Type Fixes	Bukola	11/22/25	None - correcting 0-5 scale slider behavior and styling
Cross-Browser Mobile Testing	Wylie, Bukola	11/27/25	Requires access to iOS Safari, older Android devices
Dependabot Security Vulnerabilities	Wylie, Vitaliy	11/30/25	Yes - Critical: form-data unsafe random function (Issue #11)

Planned Work for Next Period

Task	Owner	Planned Start	Target Completion
Front end work	Bukola and Denia	11/15/25	ASAP
General bug fixes	Wylie and Vitaliy	11/15/25	ASAP
Optimize Backend API	Wylie	11/24/25	11/30/2025
Survey Welcome & Thank You Pages	Bukola, Denia	11/25/25	12/02/25
Survey Start Animation	Denia	11/28/25	12/05/25
Migration: Hardcoded Questions to Database	Vitaliy	11/20/25	12/01/25

Issues

Survey Creator/Management Feature	Vitaliy, Wylie	12/02/25	12/15/25
TypeScript Migration Assessment	Wylie	11/22/25	11/29/25
Load Testing (100+ Concurrent Users)	Wylie, Vitaliy	12/03/25	12/08/25
Accessibility Audit (WCAG 2.1 AA)	Bukola	12/05/25	12/12/25
Firebase Backup Strategy Configuration	Vitaliy	11/25/25	12/01/25
Error Monitoring Setup (Sentry)	Wylie	11/27/25	12/03/25
QR Code Generation for Surveys	Vitaliy	12/06/25	12/10/25
Zip Code Heatmap Visualization	Wylie, Vitaliy	12/08/25	12/15/25
Multilingual Support (Spanish)	Denia, Bukola	12/09/25	ASAP/2026 Q1
Social Media Brand Assets	Denia	12/10/25	12/13/25
Stakeholder Demo Preparation	All Team	12/12/25	12/15/25
Security Vulnerability Remediation	Wylie, Vitaliy	11/18/25	11/30/25

LogDocuments and monitors issues, assigned resolutions, and current statuses.

Issue ID	Description	Priority	Status	Resolution	Date Resolved	Owner
001	API response delay	High	In Progress	Optimizing queries	TBD	Wylie
002	Frontend not rendering on deployment. Localhost works as intended. Vercel & Netlify deploy successfully to a white static page. Vercel logs do not prompt even an	High	In Progress	Not found	TBD	Vitaliy/Wylie

	ignorable warning. All green checks. Inspecting the page w/ dev tools shows 1 backend misconfig warning, but no conclusive errors and backend works.					
005	Partial survey submission UX decision needed	Medium	Blocked	Awaiting stakeholder feedback on data integrity vs. completion rates	TBD	Bukola, Denia

Change Control Log

Tracks and evaluates changes made during execution. Impact (High, Medium, Low)

Change ID	Description	Impact	Decision	Date	Owner
001	Deploy on Google Cloud instead of AWS	High	Deployed to GCP	11/10/25	Wylie
002	MaterialUI Implementation				
003					

LESSONS LEARNED REPORT

1. What areas of the project did the team do well?

The initial sprint consisted of strong, consistent communication that followed best practices. We successfully reproduced the original team's POC within the first sprint, providing a solid foundation for development. Technically, our burndown rate increased per sprint following this. Our team recognized early completion of Sprint 2 mobile app tasks from POC deployment, and began Sprint 3 admin dashboard development ahead of schedule, delivering more features than originally planned. In essence, proactively expanded scope. We created extensive technical documentation that is accurate, clear and once diligently organized, will benefit future teams and stakeholders. Database creation, migration, and set-up for data intake/processing was incredibly well done.

2. What areas of the project did the team **not** do well and why?

Sprint 2 and Sprint 3 tasks overlapped due to an accelerated timeline, making it challenging to track actual sprint boundaries in retrospectives. As a consequence, documentation and formal submission deliverables proved to be difficult to create. Additionally, UI/UX is a far cry from brand consistency for VAI, The VAI branding (logo, colors, typography) were not implemented in time for live demo as planned, requiring retroactive updates to existing components in the next sprint.

3. How can the team improve its performance?

All members must have better time management, and exercise their agency to take ownership of chunks of the project to ensure responsibility is felt, not merely seen as a deadline. Current devs need to optimize the steps required for other members to contribute components. Everyone should be able to describe the system's architecture, core components, and how they are related.

PARTICIPATION REPORT

List team member participation during **reporting period**

Guidelines for Reporting Participation:

1. Total Tasks Assigned – Count all tasks assigned to the member.
2. Tasks Not Started or Delayed – Total tasks that not yet started missed deadlines or have blockers.
3. Tasks In Progress – Total tasks actively being worked on.
4. Tasks Completed – Total number of successfully completed tasks.
5. % of Work Completed – $(\text{Tasks Completed} \div \text{Total Tasks Assigned}) \times 100\%$
6. % of Team Workload – $(\text{Total Tasks Assigned} \div \text{Total Tasks for Team}) \times 100\%$

Team Member: Name & Role	Total Tasks Assigne d	Tasks Not Started or Delayed	Tasks In- Progress	Tasks Complete d	% of Work Complete d	% of Team Workload	Notes
Bukola Famurewa	2	1	1	0	40%	10%	Still working on Color Scheme to use for the admin dashboard
Vitaliy Dmukhovskyy	-	<10	5 tasks as of 11/15; due 11/23	-	<i>Estimate: 75%</i>	45%	Working on the demo takes a lot of time and set up database is difficult
Wylie Kuhn	2	1	1	1	80%	30%	Working on the demo takes a lot of time and set up database is difficult
Denia Casimiro	1	1	1	1	100%	5%	
Jadon Tay	1	0	0	1	100%	10%	Filling out the forms as best as i can
Total						100%	

7. Notes/Comments – Provide explanations, blockers, or additional details.

PRESENTATION & PEER REVIEW

Presentation

Present the overall project and analysis of the new system. Include the following:

- Introduction of the team
- Give an overview of the project
- Walkthrough the Project Execution/Demo of any working system component
- Wrap up the presentation
- Q&A

Peer Review

Review and rate all other teams' projects, products, and presentations using the criteria below.

- Submit a constructive review that explains each rating.
- Highlight strengths, weaknesses, and suggestions for improvement.

Rating Scale: 1 = Poor, 2 = Unsatisfactory, 3 = Average, 4 = Good, 5 = Excellent

Criteria:

- Project Quality: Meets project management standards
- Product Quality: Meets product quality standards
- Presentation:
 - o Content: Accurate, relevant, and complete for the audience
 - o Organization: Logically structured and cohesive
 - o Delivery: Clear audio/visuals, understandable communication
 - o Length: Fits within time limits
 - o Subject Matter Proficiency: Presenters demonstrate expertise through responses

Peer Review Feedback

Team	What the Team did well	What the Team did NOT do well	How to improve	Project Rating	Product Rating	Presentation Rating	Total
1.	N/A as going on monday	N/A as going on monday	N/A as going on monday	N/A	N/A	N/A	N/A
2.	Good presentation has graphs	n/a	Actually show off the demo instead of a video	5	5	5	15
3.	Like the fonts and looks pretty	Presentation lack picture	Video or live demonstration	5	5	5	15
4.	Good presentations	Needs more color	Add color	5	5	5	15
5.	Good presentation as it is short and concise	No demo for the moment	More effort into the slides	4	5	4.5	13.5
6.							
7.							
8.							

Peer Review Sign-off

Completed by Name(s):	Vitaliy Dmukhovskyy, Jadon Tay
Signature(s):	V.D.K, Jadon Tay
Date Completed:	11/11/2025
Group Number:	6

Works Cited

- Kitsaras, George, et al. "An Interactive Text Message Survey as a Novel Assessment for Bedtime Routines in Public Health Research: Observational Study." *JMIR Public Health and Surveillance*, vol. 6, no. 4, 2020, article e15524. *PubMed Central*, <https://doi.org/10.2196/15524>.
- Nova Scholar Team. "The Comprehensive Guide to Data Gathering in Psychology." *Nova Scholar*, 25 Oct. 2025, www.novascholar.org/blog-posts/the-comprehensive-guide-to-data-gathering-in-psychology. Accessed 17 Nov. 2025.
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- "Branding." McKinsey & Company, McKinsey, 2025, www.mckinsey.com/capabilities/growth-marketing-and-sales/how-we-help-clients/branding.
- "Adopting an Ecosystem View of Business Technology." McKinsey & Company, McKinsey, 13 Feb. 2017, www.mckinsey.com/capabilities/mckinsey-digital/our-insights/adopting-an-ecosystem-view-of-business-technology.
- "Digital Community Engagement Case Studies." Local Government Association, 2024, www.local.gov.uk/pas/plan-making/case-studies/digital-community-engagement-case-studies.
- "Case Study: Building an Equitable Community Engagement Practice." City of Philadelphia, 25 Apr. 2022, www.phila.gov/2022-04-25-case-study-building-an-equitable-community-engagement-practice/. Updated 2025.
- "From Community Engagement to Ownership: Tools and Case Studies." Urban Sustainability Directors Network, 2023, www.usdn.org/uploads/cms/documents/community_engagement_to_ownership_-_tools_and_case_studies_final.pdf.

Original POC Repository: [iden-a/VAI-DataSystem](https://github.com/iden-a/VAI-DataSystem)

GitHub: <https://github.com/iden-a/VAI-DataSystem> Referenced for initial architecture, wireframes, and survey question design.

Van Alen Institute Official Website: <https://www.vanalen.org>

Primary reference for ongoing brand consistency guidelines (logo, typography, color palette) -- Next sprint addresses full branding implementation, using ongoing Figma collaboration —

Firestore Documentation

Firestore Authentication: <https://firebase.google.com/docs/auth>

Firestore Database: <https://firebase.google.com/docs/firestore>

Firestore Admin SDK:

<https://firebase.google.com/docs/admin/setup>

Firestore Costs: <https://cloud.google.com/firestore/pricing>

Google Cloud Run Documentation

Deployment Guide:

<https://cloud.google.com/run/docs/quickstarts/build-and-deploy/deploy-python-service>

Container Registry:

<https://cloud.google.com/container-registry/docs>

Cloud Compute Costs <https://cloud.google.com/run/pricing>

Vercel Documentation:

Deployment Guide: <https://vercel.com/docs>

Costs: <https://vercel.com/pricing>

React Documentation

React 19 Docs: <https://react.dev>

React Router v7: <https://reactrouter.com/en/main>

Chart.js Documentation

Chart.js v4: <https://www.chartjs.org/docs/latest/>

chartjs-plugin-datalabels: <https://chartjs-plugin-datalabels.netlify.app>

Material UI Documentation

MUI v6: <https://mui.com/material-ui/getting-started/>

Technology Stack References

Technology	Version	Documentation URL
React	19.1.0	https://react.dev
Vite	6.0.11	https://vite.dev
Flask	3.1.0	https://flask.palletsprojects.com
Firestore Admin SDK	6.8.0	https://firebase.google.com/docs/admin/setup
Chart.js	4.4.9	https://www.chartjs.org
pandas	2.2.3	https://pandas.pydata.org
matplotlib	3.10.1	https://matplotlib.org

AI Tool	Task Category	Specific Use Case	Justification	Member
GitHub Copilot	Material UI Research	Researched best practices for Material UI theming, component composition, and responsive design patterns	Accelerated learning curve for MUI v6 components and ensured consistent styling across the dashboard. Debating the approach and ultimately determined to append the backlog to include MUI v6 component implementation.	Vitaliy
GitHub Copilot	CSS Token Management	Reviewed and validated CSS custom properties and tokens during MaterialUI implementation	Ensured reusable design system tokens and prevented hardcoded style values; opened up future UI/UX potential	Vitaliy
GitHub Copilot	Material UI Testing	Provided guidance on testing Material UI components with React Testing Library and Jest	Improved test coverage for complex UI interactions (modals, charts, filters)	Vitaliy
Claude Sonnet 4.5	Documentation Generation	Generated PHASE-TWO-CHANGELOG.md by analyzing git commit history and summarizing feature implementations with line counts	Saved ~6 hours of manual documentation work; ensured accurate feature tracking for stakeholder reporting	Vitaliy
Claude Sonnet 4.5	Dependency Vulnerability Analysis	Reviewed Dependabot security alerts and recommended manual fixes for critical vulnerabilities in Flask and React dependencies	Ensured production-ready security posture before live demo; prioritized critical vs. low-severity updates	Vitaliy

Claude Sonnet 4.5	Codebase Review	Prompt: As a technical expert, analyze this codebase and provide a clear project overview: - Core Components: Describe the major components or modules, their responsibilities, and any key classes or functions they contain. Note any relevant design patterns or architectural approaches. - Component Interactions: Explain how the components interact, including data and control flow, communication methods, and any APIs or interfaces used. Highlight use of dependency injection or service patterns if applicable. - Deployment Architecture: Summarize the deployment setup, including build steps, external dependencies, required environments (e.g., dev, staging, prod), and infrastructure or containerization details. - Runtime Behavior: Describe how the application initializes, handles requests and responses, runs business workflows, and manages errors or background tasks. Stay focused on providing a technical overview that helps a developer quickly understand how the system works. Avoid personal opinions or implementation suggestions unless specifically asked. Ask me any questions to ensure you	To create a simple, single page report to be pinned always to the root of our repository for both team members and stakeholders to glean. Understanding the basic flow of data in the context of business processes (users complete surveys for an organization) and the specific backend/frontend code that enables functionality of the system.	Vitaliy
CIS-Prof. Brown			41	

AI Tool	Task Category	Specific Use Case	Justification	Member
		are 95%+ confident before beginning.		
GitHub Copilot	Chart.js Integration	Provided examples for implementing Chart.js with custom datalabels plugin and responsive configuration	Accelerated dashboard analytics development; charts rendered correctly on first implementation	Vitaliy
Claude Sonnet 4.5	Documentation Review	Reviewed system-overview.md, SysArch.md, and MACOS-SETUP.md for clarity, completeness, and technical accuracy, using the POC (authors/parties anonymized) as baseline reference.	Structured documentation neatly in markdown to match prior team's POC repo documentation and quality standards; ensure doc is accessible to non-technical stakeholders	Vitaliy

Integrity Statement

By signing below, I affirm that the work submitted for this assignment is my own, was not outsourced, provided by a third party, or written using artificial intelligence, and I have complied with the college's "Academic Honesty" policies.

This project was completed in accordance with the academic integrity policies of CIS 5800 EMWA. All code was developed by the Phase Two development team, with the following acknowledgments:

The original Proof of Concept repository, developed by **iden-a** and **Michael Schlichting Barbey** and their team. Their POC offered us an inspiring, development-ready foundation to work from. Their rigorous research, initial mobile-facing app implementation, brand-consistent wireframes, and Firebase integration significantly improved our team's ramp-up time.

Code Attribution:

Any code snippets or algorithms inspired by external sources are properly cited in comments and documentation.

External Libraries:

All third-party libraries are properly credited and used under their respective open-source licenses (MIT, BSD, Apache 2.0).

#	Team Member	Project Role	Signature	Date
1.	Jadon Tay	Project Manager	Jadon Tay	11/12/25
2.	Vitaliy Dmukhovskyy	Fullstack Developer	V.D.K	11/12/2025
3.	Bukola Famurewa	UI/UX designer	Bukola Famurewa	11/14/2025
4.	Wylie Kuhn	Backend developer	Wylie Kuhn	11/14/25
5.	Denia Casimiro	Designer	Denia Casimiro	11/14/25

Appendix A: Demo Screenshots

(Screenshots from live demo on November 10, 2025 are available in the repository at [FormalDeliverableDocs/demo-screenshots/](#))

Appendix B: Sprint Planning Documents

- SprintSchedule.md (full sprint task breakdown)
- Monday.com project board export (if available)

Appendix C: Technical Architecture Diagrams

- System architecture diagram (see system-overview.md)
- Deployment architecture diagram (see MACOS-SETUP.md)
- Data flow diagrams (see SysArch.md)